

Name: _____

Date: _____

DECIMALS,

(ADDING AND SUBTRACTING DECIMALS)

LO: I can add and subtract decimals using column methods, lining up the decimal points.

Adding and subtracting decimals

When adding or subtracting **decimals**, line up the **decimal points** so that digits with the same place value are in the same column.

Tip: Use trailing zeros as placeholders so all numbers have the same number of decimal places.

Key points

●	$3.5 + 2.7 = 6.2$	line up points
●	$5.0 - 1.3 = 3.7$	use placeholder 0
●	$12.45 + 3.6 = 16.05$	$3.6 = 3.60$
●	$3.5 + 2.7 = 5.12$	not place value error
●	$1.3 + 2.45$ lined up without point	must align decimals

Key idea

Line up the decimal points. Fill empty places with zeros. Then add or subtract as normal.

$$\textcircled{12.45} + \textcircled{3.60} = \textcircled{16.05}$$

Pad with zeros to match decimal places

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - adding

Calculate $4.3 + 2.8$

Line up: $4.3 + 2.8$

$3 + 8 = 11$, write 1 carry 1

$4 + 2 + 1 = 7$ 7.1

= 7.1

Carry works exactly the same as with whole numbers.

Example 2 - different decimal places

Calculate $5.6 + 3.45$

Write as $5.60 + 3.45$

$0 + 5 = 5$; $6 + 4 = 10$, write 0 carry 1

$5 + 3 + 1 = 9$ 9.05

= 9.05

Pad 5.6 with a zero to make 5.60.

Example 3 - subtraction

Calculate $8.2 - 3.7$

Line up: $8.2 - 3.7$

$2 - 7$: borrow, $12 - 7 = 5$

$7 - 3 = 4$ 4.5

= 4.5

Borrow from the next column just like whole numbers.

Example 4 - from a whole number

Calculate $10 - 3.46$

Write as $10.00 - 3.46$

$0 - 6$: borrow through $10 - 6 = 4$

Continue borrowing: $9 - 4 = 5$, $9 - 3$

$= 6$ 6.54

= 6.54

Write 10 as 10.00 to subtract easily.

YOUR TURN – PRACTICE (deliberate practice)

1. Adding decimals (same d.p.)

Calculate each sum.

- a) $3.4 + 2.5$
- b) $6.7 + 1.8$
- c) $4.25 + 3.14$
- d) $8.63 + 5.29$

2. Adding (different d.p.)

Pad with zeros first.

- a) $4.5 + 2.35$
- b) $7 + 3.62$
- c) $12.4 + 0.85$
- d) $9.1 + 4.275$

3. Subtracting decimals

Calculate each difference.

- a) $7.8 - 3.5$
- b) $9.4 - 2.7$
- c) $6.32 - 4.18$
- d) $10 - 4.65$

4. Word problems

Show your column working.

- a) Buy items at £3.49 and £2.75. Total?
- b) Had £20; spent £13.62. Change?
- c) Run 3.5 km then 2.75 km. Total?
- d) Rope 8 m; cut off 2.35 m. Left?

5. Mixed addition and subtraction

Calculate.

- a) $5.6 + 3.45 - 2.1$
- b) $10 - 3.7 + 1.25$
- c) $7.82 + 4.5 - 6.07$
- d) $20 - 8.45 - 3.2$
- e) $0.75 + 0.8 + 0.125$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Line up decimal points when adding ☐

Correct answer: _____

Reason: _____

b) $3.5 + 2.7 = 5.12$ ☐

Correct answer: _____

Reason: _____

c) $10 - 3.46 = 6.54$ ☐

Correct answer: _____

Reason: _____

d) 4.5 is the same as 4.50 ☐

Correct answer: _____

Reason: _____

e) $1.3 + 2.45 = 3.48$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shopper buys items costing **£4.99**, **£2.50**, **£7.25** and **£0.89**. They pay with a **£20 note**. How much change do they receive?

Expression: _____

Simplified: _____

b) Three pieces of wood are **2.4 m**, **1.85 m** and **3.125 m** long. What is their total length? If they are cut from a **10 m** plank, what length remains?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

